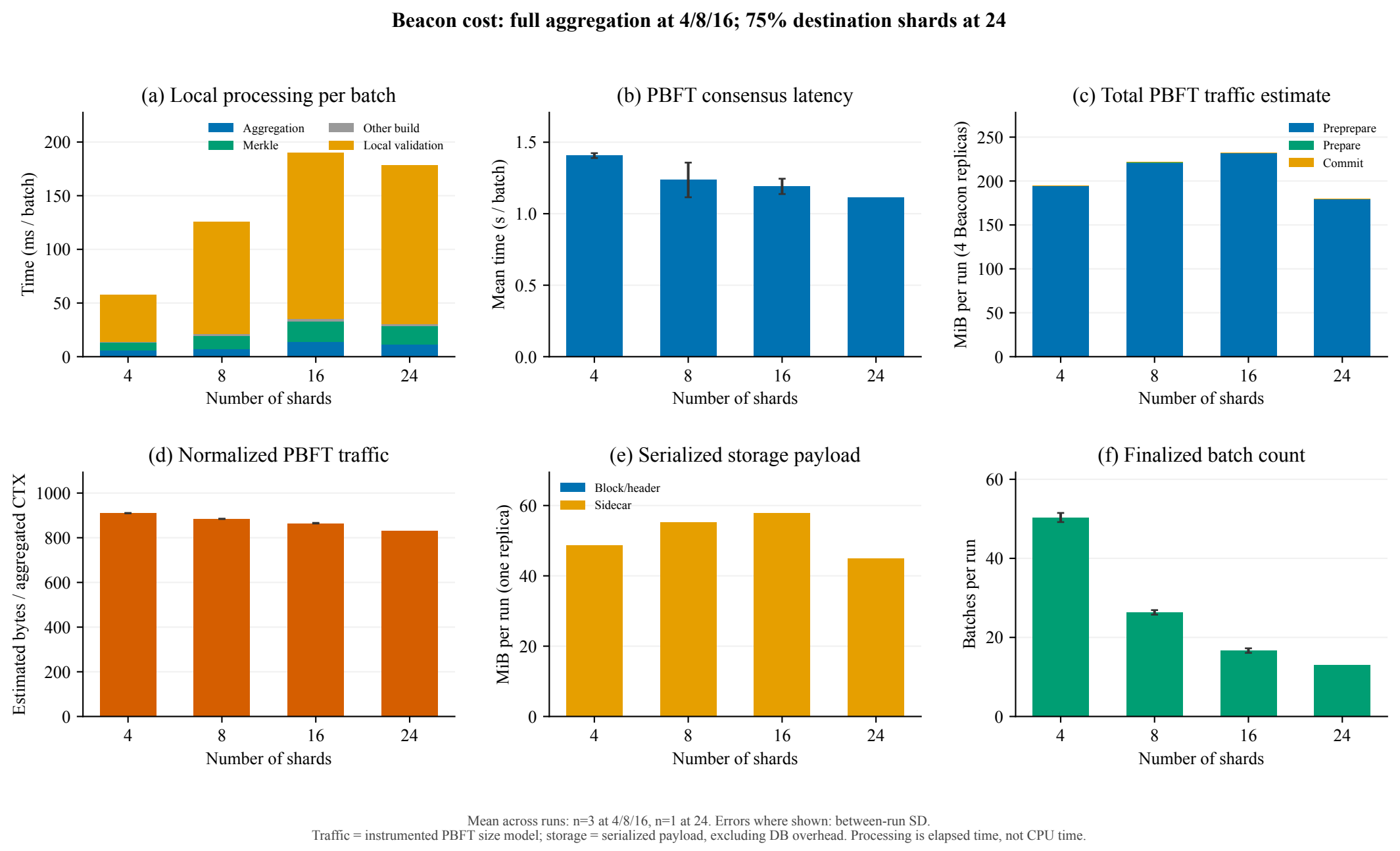

Beacon cost: full aggregation at 4/8/16; 75% destination shards at 24
(a) Local processing per batch
Aggregation
Other build
Merkle
Local validation
Time (ms / batch)
Number of shards
(b) PBFT consensus latency
Mean time (s / batch)
Number of shards
(c) Total PBFT traffic estimate
Preprepare
Prepare
Commit
MiB per run (4 Beacon replicas)
Number of shards
(d) Normalized PBFT traffic
Estimated bytes / aggregated CTX
Number of shards
(e) Serialized storage payload
Block/header
Sidecar
MiB per run (one replica)
Number of shards
(f) Finalized batch count
Batches per run
Number of shards
Mean across runs: n=3 at 4/8/16, n=1 at 24. Errors where shown: between-run SD.
Traffic = instrumented PBFT size model; storage = serialized payload, excluding DB overhead. Processing is elapsed time, not CPU time.